

Green Bond Thesis — Data Source Documentation

Title: Do Politicians' Words Have a Dollar Value?

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This document provides a complete inventory of all data sources, sample sizes, and coverage dates used in the thesis. It serves as a reference for the Data chapter (Chapter 3) and is required for supervisor submission.

1. Financial Data — Bond Yields

Source: Refinitiv (via CEU institutional access)

Format: Daily Bid/Ask yields in .xlsx files

Processing: $\text{Mid-YTM} = (\text{BidYld} + \text{AskYld}) / 2$

India — Green Bonds

Bond	Type	ISIN	Currency	Issuance	Maturity	Start	End	Daily Obs
India Green 10Y	Sovereign Green Bond	IN0020220144	INR	Jan 2023	~2033	27-Jan-2023	20-Apr-2026	842
India Green 5Y	Sovereign Green Bond	IN0020230143	INR	Nov 2023	~2028	13-Nov-2023	20-Apr-2026	634

India — Conventional Twins

Bond	Type	ISIN	Currency	Issuance	Maturity	Start	End	Daily Obs
India Conv 10Y	GOI Fixed Coupon	IN0020190362	INR	2019	Oct 2029	01-Jan-2020	21-Apr-2026	1,645

Bond	Type	ISIN	Currency	Issuance	Maturity	Start	End	Daily Obs
India Conv 5Y	GOI Fixed Coupon (7.26%)	IN072629G	INR	~2019	~2029	01-Jan-2020	21-Apr-2026	1,645

Indonesia — Green Bonds

Bond	Type	ISIN	Currency	Issuance	Maturity	Start	End	Daily Obs
Indonesia Green 10Y	Sovereign Green Sukuk	US71567RAV87	USD	Jun 2022	~2032	06-Jun-2022	20-Apr-2026	1,011
Indonesia Green 2033	Sovereign Green Sukuk	US71567RAY27	USD	Nov 2023	Nov 2033	15-Nov-2023	20-Apr-2026	634

Indonesia — Conventional Twins

Bond	Type	ISIN	Currency	Issuance	Maturity	Start	End	Daily Obs
Indonesia Conv 10Y	Republic of Indonesia	US455780DN36	USD	Sep 2022	~2032	20-Sep-2022	20-Apr-2026	935
Indonesia Conv 2033	Republic of Indonesia (4.85%)	US455780DR40	USD	Jan 2023	Jan 2033	11-Jan-2023	20-Apr-2026	854

Bond Data Summary

Metric	Value
Total bonds	8 (4 green + 4 conventional)
Total daily observations	8,200
Countries	India, Indonesia
Currencies	INR (India), USD (Indonesia)
Source platform	Refinitiv Workspace

2. Control Variables

These variables control for global and country-specific financial conditions that affect bond yields independently of climate sentiment.

Variable	Description	Source	RIC / Ticker	Currency	Start	End	Daily Obs
VIX	CBOE Volatility Index	FRED (St. Louis Fed)	VIXCLS	USD	02-Jan-2020	16-Apr-2026	1,641
INR/USD	US Dollar / Indian Rupee spot rate	Refinitiv	INR=	INR per USD	01-Jan-2020	20-Apr-2026	1,644
IDR/USD	US Dollar / Indonesian Rupiah spot rate	Refinitiv	IDR=	IDR per USD	01-Jan-2020	20-Apr-2026	1,644
India CDS 5Y	India sovereign 5-year CDS spread	Investing.com	INGV5YUSAC=R	BPS (USD)	01-Jan-2020	26-Sep-2025	856
Indonesia CDS 5Y	Indonesia sovereign 5-year CDS spread	Investing.com	IDGV5YUSAC=R	BPS (USD)	01-Jan-2020	17-Apr-2026	1,622

Control Variables Summary

Metric	Value
Total variables	5
Total observations	7,407
Sources	FRED, Refinitiv, Investing.com

Role in Regression

Variable	Transformation	Purpose
VIX	log-transformed, first-differenced	Controls for global risk appetite shocks
FX Rate	log-transformed, first-differenced, country-specific	Controls for currency depreciation effects on yield spreads
CDS Spread	first-differenced, country-specific	Controls for sovereign default risk

Data Limitations

- **India CDS** ends 26-Sep-2025 — earlier than other variables. This restricts the India panel to Jul-2024 when combined with the complete-case restriction.
- **VIX** is forward-filled over weekends and holidays when bond markets are closed but VIX is not quoted.

3. Sentiment Data — GDELT V2Tone (Primary Measure)

Source: GDELT 2.0 Global Knowledge Graph (GKG)
Access method: Google BigQuery (public dataset `gdeltd-bq.gdeltdv2.gkg`)
Processing notebook: `GreenBond_GDELT_Clean.ipynb` (Google Colab, T4 GPU)

Pipeline

Step	Description	Output
1	Query GDELT GKG via BigQuery for articles mentioning India/Indonesia + climate themes	Raw articles
2	Sample 1,000 articles per country per month for computational feasibility	Sampled articles
3	Parse V2Tone field: normalise to [-1, +1]	Article-level sentiment
4	Aggregate to daily country-level mean and standard deviation	Daily sentiment series

GDELT Query Filters

Filter	Value
Date range	20200101000000 – 20241231235959 (14-digit GDELT format)
Country geocodes	<code>India#IN</code> , <code>Indonesia#ID</code>
Theme filter	<code>CLIMATE%</code> , <code>ENV_%</code> , <code>ENERGY%</code> , <code>GREEN%</code>

Output Variables

Variable	Description	Scale	Obs
<code>sentiment_gdeltd</code>	Mean daily V2Tone across all climate-related articles	[-1, +1]	3,654 country-days
<code>sentiment_std</code> (Signaling Noise)	Standard deviation of article-level V2Tone within each day	[0, +∞)	3,654 country-days

Variable	Description	Scale	Obs
article_count	Number of articles per country-day	Integer	3,654 country-days
pct_positive	Proportion of articles with positive V2Tone	[0, 1]	3,654 country-days
pct_negative	Proportion of articles with negative V2Tone	[0, 1]	3,654 country-days

GDELT Sentiment Summary

Country	Days	Mean Sentiment	Mean Signaling Noise	Total Articles
India	1,827	-0.063	0.335	60,000
Indonesia	1,827	-0.087	0.402	60,000
Total	3,654			120,000

Key Design Decision: V2Tone as Primary Sentiment Measure

GDELT V2Tone is a pre-computed sentiment score embedded in every GKG record. It measures the overall tone of an article on a continuous scale, where positive values indicate positive tone and negative values indicate negative tone. The score is normalised to [-1, +1] by dividing by 10 and clipping.

Signaling Noise is operationalised as the daily standard deviation of article-level V2Tone scores. High sentiment_std on a given day means climate news coverage was internally contradictory — some articles positive, some negative — which is the thesis's measure of the Credibility Gap.

4. Sentiment Data — ClimateBERT (Robustness Check)

Source: GDELT DOC API → ClimateBERT (HuggingFace)
Access method: Google Colab (T4 GPU)
Processing notebook: GreenBond_ClimateBERT_clean.ipynb

Why ClimateBERT?

ClimateBERT (Webersinke et al., 2022) is a transformer-based NLP model trained specifically on climate-related disclosures. Unlike V2Tone which measures general media tone,

ClimateBERT is designed to distinguish between vague environmental rhetoric and credible policy commitments.

Two-Stage Pipeline

Stage	Model	Purpose	Labels
1 — Detection	distilroberta-base-climate-detector	Filter to climate-relevant paragraphs	yes / no
2 — Sentiment	distilroberta-base-climate-sentiment	Classify sentiment of climate paragraphs	opportunity (+1) / neutral (0) / risk (−1)

Important: ClimateBERT is trained on **paragraphs**, not single sentences. Headlines from the same country-day are concatenated into a single paragraph before scoring.

Data Collection: GDELT DOC API

Parameter	Value
Query (India)	climate India government
Query (Indonesia)	climate Indonesia government
Time windows	10 half-year periods (2020-H1 through 2024-H2)
Max records per request	250
Rate limit handling	Exponential backoff (30s → 60s → 120s → 240s)

Collection Results

Metric	Value
Total articles fetched	4,891
English articles (after language filter)	1,539
After deduplication	1,539
India articles	992
Indonesia articles	547

Scoring Results

Metric	Value
Daily paragraphs constructed	70 country-days
Climate-relevant (Stage 1)	54 / 70 (77.1%)

Metric	Value
Label distribution	opportunity: 28, neutral: 16, risk: 10
India mean score	+0.294
Indonesia mean score	+0.351

Validation: V2Tone vs ClimateBERT Correlation

Country	Correlation (r)	Matched Days	Interpretation
India	+0.276	16	Weak positive — directionally consistent
Indonesia	−0.037	36	Near zero — measures diverge
Overall	+0.035	52	Weak overall agreement

ClimateBERT Limitations

Limitation	Detail
Sparse coverage	Only 52 matched days out of 3,654 total (1.4%) due to API constraints
First-differenced subsample	Only 20 observations survive first-differencing — too few for reliable regression
Paragraph-level scoring	Model trained on corporate disclosures, not political news headlines
Label mapping	Model outputs opportunity/risk rather than positive/negative — mapped as equivalent
No pre-training on Asian political speech	Model validated on English-language corporate disclosures, not Hindi/Indonesian political media

5. Treatment Events for DiD / Event Study

Country	Event	Date	Description	Source
India	Sovereign Green Bond Auction	25-Jan-2023	India's inaugural green bond issuance (₹160 billion, two tranches: 5Y and 10Y)	Ministry of Finance India
Indonesia	JETP Announcement	15-Nov-2022	Just Energy Transition Partnership announced at G20 Bali Summit (\$20 billion mobilised)	G20 Indonesia Presidency

6. Regression Panel — Final Sample

The regression panel is constructed by merging bond yield data, sentiment, and all control variables, then restricting to complete cases within 2022–2024.

Panel Construction Steps

Step	Action	Resulting Obs
1	Compute Greenium for 4 twin-bond pairs	3,045
2	Merge sentiment (country-specific)	3,045
3	Merge VIX, FX, CDS	3,045
4	Restrict to $\leq$ Dec 2024	~2,500
5	Drop incomplete cases (missing sentiment, CDS, FX)	1,455
6	First-differencing (lose 4 first-obs per pair)	1,451

Final Panel by Country and Tenor

Country	Tenor	Obs (levels)	Obs (first-diff)	Start	End	Mean Greenium
India	10Y	386	385	27-Jan-2023	19-Jul-2024	−0.015 pp
India	5Y	178	177	15-Nov-2023	19-Jul-2024	−0.028 pp
Indonesia	10Y	596	595	20-Sep-2022	31-Dec-2024	−0.069 pp
Indonesia	2033	295	294	15-Nov-2023	31-Dec-2024	+0.036 pp
Total		1,455	1,451			

Country Subsets for DiD

Country	Obs (first-diff)	Pairs	Period
India	562	India_10Y, India_5Y	Jan-2023 – Jul-2024
Indonesia	889	Indonesia_10Y, Indonesia_2033	Sep-2022 – Dec-2024

Event Study Samples (± 10 weeks around treatment)

Sample	Obs	Pre-event weeks	Post-event weeks
India	55	0 (no pre-treatment bond data)	11

Sample	Obs		Pre-event weeks	Post-event weeks
Indonesia	105	10		11
Pooled	160	10		11

7. Data Source Summary

All Sources at a Glance

Data Category	Source	Access Method	Observations
Bond yields (8 bonds)	Refinitiv	CEU institutional login	8,200 daily
VIX	FRED (St. Louis Fed)	Public download	1,641 daily
Exchange rates (INR, IDR)	Refinitiv	CEU institutional login	3,288 daily
CDS spreads (India, Indonesia)	Investing.com	Public download	2,478 daily
GDELT V2Tone sentiment	GDELT GKG via BigQuery	Google Cloud project	120,000 articles → 3,654 country-days
ClimateBERT sentiment	GDELT DOC API + HuggingFace	Google Colab (T4 GPU)	1,539 articles → 52 matched days
Total unique data sources	5		

File Inventory

File	Location
IND_GREEN_10Y_IN0020220144_27Jan2023_20Apr2026.xlsx	Bond_Yield_Data/india_bond/
IND_GREEN_5Y_IN0020230143_13Nov2023_20Apr2026.xlsx	Bond_Yield_Data/india_bond/
IND_CONV_10Y_IN0020190362_01Jan2020_21Apr2026.xlsx	Bond_Yield_Data/india_bond/
IND_CONV_5Y_IN072629G_01Jan2020_21Apr2026.xlsx	Bond_Yield_Data/india_bond/
IDO_GREEN_10Y_US71567RAV87_06Jun2022_20Apr2026.xlsx	Bond_Yield_Data/indonesia_bond/
IDO_GREEN_2033_US71567RAY27_15Nov2023_20Apr2026.xlsx	Bond_Yield_Data/indonesia_bond/
IDO_CONV_10Y_US455780DN36_20Sep2022_20Apr2026.xlsx	Bond_Yield_Data/indonesia_bond/
IDO_CONV_2033_US455780DR40_11Jan2023_20Apr2026.xlsx	Bond_Yield_Data/indonesia_bond/

File	Location
VIXCLS.csv	Control_Variables/VIX/
CTRL_USDINR_INR_01Jan2020_20Apr2026.xlsx	Control_Variables/Exchange_R
CTRL_USDIDR_IDR_01Jan2020_20Apr2026.xlsx	Control_Variables/Exchange_R
CTRL_CDS5Y_India_INGV5YUSAC_01Jan2020_26Sep2025.csv	Control_Variables/CDS/
CTRL_CDS5Y_Indonesia_IDGV5YUSAC_01Jan2020_17Apr2026.csv	Control_Variables/CDS/
daily_sentiment_merged.csv	GDELT_data/
regression_panel.csv	Output/

Key Data Limitations

Limitation	Impact	Mitigation
India CDS ends Sep-2025	India panel restricted to Jul-2024	Acknowledged in limitations; does not affect treatment period
Indonesia conventional twins start post-2022	Limited pre-treatment data for Indonesia	Parallel trends satisfied by construction (Zerbib 2019 matching design)
India bonds issued at/after treatment date	Cannot formally test parallel trends for India	Twin-bond matching design substitutes for empirical parallel trends test
GDELT sentiment ends Dec-2024	Panel truncated at Dec-2024 despite bond data extending to Apr-2026	Treatment events (Nov 2022, Jan 2023) are well within coverage window
ClimateBERT coverage: 52 days / 3,654 (1.4%)	ClimateBERT cannot serve as primary variable	Used as robustness check only; V2Tone is primary sentiment measure
ClimateBERT all-zero on theme codes	Original pipeline failure	Fixed by fetching real headlines via GDELT DOC API + two-stage scoring